Fran Hsu

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Data science graduate student with 5 years of consulting experience transforming complex transportation, infrastructure, and operational challenges into data-driven solutions. Experienced in bridging technical analysis with business requirements to support real-world decision-making.

Education

University of Pittsburgh Expected Graduation: May 2027
M.S. in Information Science, GPA: 4.0 Pittsburgh, PA
- Relevant Courses: Database Management, Machine Learning, Human Information Processing

National Cheng Kung University Sep 2016 – Jun 2018
M.S. in Transportation and Communication Management Science, GPA: 3.69 Tainan, Taiwan

Work Experience

University of Pittsburgh, Katz Graduate School of Business Aug 2026 – Present
Engagement Manager, Consulting Field Project — PITT OHIO / LogiFlow
- Lead a 5-person consulting team assessing growth opportunities for a customizable transportation management system (TMS), identifying target markets beyond its industry-concentrated customer base
- Identified 3 priority customer segments through analysis of historical customer data, market sizing (TAM/SAM/SOM), and benchmarking of 15 competitors to inform TMS growth strategy

GEOINT-ISAC (Geospatial Intelligence Information Sharing and Analysis Center) May 2026 – Aug 2026
Data & Intelligence Intern
- Structured climate-risk data for a climate-risk atlas covering six U.S. military installations, validated U.S. site coordinates, and processed climate projections in Python to support GIS integration and identify data gaps
- Authored a white paper chapter synthesizing international extreme-weather case studies and co-led an 18-person team, using a progress checklist to coordinate emergency-resource database contributions

THI Consultants, Inc. July 2020 - July 2025
Planner, Traffic Engineering and Data Analytics
- Applied established forecasting methods to estimate cargo volumes across Taiwan's major ports, incorporating historical cargo data, population growth, and economic indicators to support long-term infrastructure planning
- Led development of an automated reporting solution that integrated open transportation APIs, transformed operational data, and generated dashboards, charts, maps, and report summaries, reducing manual reporting effort by 80%
- Analyzed large-scale crash and traffic datasets to identify risk patterns and contributing factors, supporting dashboard adoption and data-driven safety improvements across 10+ local governments
- Contributed to recommendations for defining autonomous-vehicle operational design domains (ODDs), combining scenario discussions with the HD mapping team and microscopic traffic simulation to assess potential issues before deployment

Skills

- **Programming & Databases:** Python (Pandas, NumPy, Matplotlib, Scikit-learn, GeoPandas, PyTorch), SQL, MySQL, R
- **Machine Learning & Analytics:** Machine Learning, Deep Learning, Model Evaluation, Forecasting, EDA
- **Data Processing & Visualization:** Data Cleaning, Data Validation, Tableau, Power BI

Selected Projects

Multi-Objective Hazmat Transportation Route Optimization 2018
- Implemented the NSGA-II evolutionary algorithm in Python to optimize hazardous-material routes across cost, accident risk, and emergency-response objectives
- Generated and visualized Pareto-optimal alternatives, enabling decision makers to evaluate trade-offs among operational efficiency, safety, and emergency preparedness

SmartCart — Grocery Price Comparison Web Application 2025 - 2026
- Designed a normalized 15-table MySQL database with indexes, transactions, functions, and triggers to support product matching, price history, and spending analytics
- Developed Python data pipelines to clean and standardize product information, resolving inconsistent units across retailers to enable accurate unit-price comparisons